

SAT UZBEKISTAN

Telegram @satuzbhunters

Mock Test
Time 70 minutes

1.

In the equation $3x^2 - 16x = -20$, what is one possible value of x ?

2.

What number divided by two is equal to that same number minus 15?

3.

Circle O (not shown) is divided into three sectors. Points P , Q , and R are on the circumference of the circle. Sector POR has an area of 8π , and sector ROQ has an area of 6π . If the radius of circle O is 4, what is the measure of the central angle of sector QOP , in degrees?

- A) 45
- B) 90
- C) 135
- D) 180

4.

If $8x + 8y = 18$ and $x^2 - y^2 = -\frac{3}{8}$, what is the value of $2x - 2y$?

- A) $-\frac{1}{3}$
- B) $-\frac{1}{6}$
- C) $\frac{1}{3}$
- D) $\frac{1}{6}$

5.

If $a = \frac{4a^2}{16}$ and a is a nonzero integer, which of the following is equivalent to a ?

A) $4a$

B) $4\sqrt{a}$

C) $\sqrt{2a}$

D) $2\sqrt{a}$

6.

$$17s + 20t = 59$$

$$30s + 40t = 110$$

In the system of equations above, what is the value of t in terms of s ?

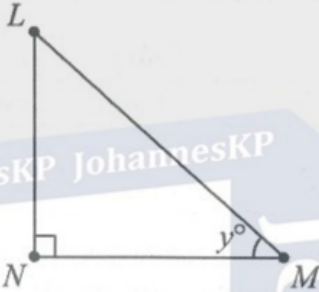
A) $\frac{2s}{5}$

B) $\frac{s}{5}$

C) $\frac{5}{2s}$

D) $\frac{5}{s}$

7.



In the figure above, if $y = 40$ and $\overline{LN} = 8$, which of the following most closely approximates the length of $\overline{MN}$?

A) 0.10

B) 9.53

C) 10.44

D) 12.45

8.

What is the value of $(2x^2 + 4x + 8) - (2x^2 - 4x + 7)$?

- A) $4x^2 + 8x + 15$
- B) $2x^2 + x + 1$
- C) $8x + 1$
- D) $8x + 15$

9.

Which of the following is equivalent to the expression $x^4 - x^3 - x^2$?

- A) $x(x^2 - x - 1)$
- B) $x(x - x^2 - x^3)$
- C) $x(x^3 - x^2)$
- D) $x^2(x^2 - x - 1)$

10.

In the inequality $37 \leq -2x + 1$, what is the appropriate order of steps needed to solve the inequality for x ?

- A) Add 1 to both sides, divide both sides by 2, and flip the inequality sign to $\geq$.
- B) Subtract 1 from both sides, divide both sides by -2 , and flip the inequality sign to $\geq$.
- C) Add 1 to both sides, divide both sides by -2 , and keep the original inequality sign.
- D) Subtract 1 from both sides, divide both sides by 2, and keep the original inequality sign.

11.

Officer Blake drives his squad car 1 mile per minute while patrolling local highways during his shift. If he has driven 480 miles by the end of his shift, how many total hours did he drive his car at the above rate?

- A) 8
- B) 12
- C) 16
- D) 20

12.

Merry joined an online community that charges a monthly fee of \$15. A one-time enrollment fee of \$50 was charged when she joined. Which of the following represents the total amount of fees that Merry has paid to the community organizers after m months, in dollars?

- A) $15m + 50$
- B) $15 + 50m$
- C) $15m - 50$
- D) $(15 + 50)m$

13.

Given that the equation $3x^2 + 2x - 8 = 0$ has two distinct solutions, what is the value of the smaller solution subtracted from the larger solution?

14.

A standard parabola in the x,y -coordinate plane intersects the x -axis at $(5, 0)$ and $(-5, 0)$. What is the value of the x -coordinate of this parabola's line of symmetry?

15.

If twice a number is equal to that number minus five, what is three times that number plus seventeen minus that number?

16.

If $2\sqrt{x} = x - 3$, which of the following is the solution set for x ?

- A) $\{-1, 9\}$
- B) $\{1, -9\}$
- C) $\{9\}$
- D) $\{1, 9\}$

17.

Cuthbert is conducting a chemistry experiment that calls for a number of chemicals to be mixed in various quantities. The one amount of which he is unsure is grams of potassium, p . If Cuthbert is certain that $(3p^2 + 14p + 24) - 2(p^2 + 7p + 20) = 0$, what is one possible value of $3p + 6$, the exact number of grams of potassium that Cuthbert would like to use for this experiment?

- A) 20
- B) 18
- C) 12
- D) 10

18.

What is the value of $(2 + 8i)(1 - 4i) - (3 - 2i)(6 + 4i)$?

(Note: $i = \sqrt{-1}$)

- A) 8
- B) 26
- C) 34
- D) 50

19.

If $x^2 + 2xy + y^2 = 64$ and $y - x = 12$, which of the following could be the value of x ?

- A) -10
- B) -4
- C) 2
- D) 10

20.

Steven needs to buy t theme park tickets for himself and his family. Each ticket costs \$80, and the number of tickets he needs to buy can be modeled by the expression $t^2 - 4t - 90 = 6$ when $t > 0$. What is the total cost of the theme park tickets that Steven purchased?

- A) \$640
- B) \$800
- C) \$960
- D) \$1,120

21.

Given the equation $y = 3x^2 + 4$, what is the function of the coefficient of 3 ?

- A) It moves the graph of $y = 3x^2 + 4$ three units higher than the graph of $y = x^2 + 4$.
- B) It moves the graph of $y = 3x^2 + 4$ three units lower than the graph of $y = x^2 + 4$.
- C) It makes the graph of $y = 3x^2 + 4$ wider than the graph of $y = x^2 + 4$.
- D) It makes the graph of $y = 3x^2 + 4$ narrower than the graph of $y = x^2 + 4$.

22.

$$2c + 3d = 17$$

$$6c + 5d = 39$$

In the system of linear equations above, what is the value of $4c - 4d$?

- A) -4
- B) 1
- C) 4
- D) 13

23.

If $3a + 2b = 24$ and $4a + 5b = 53$, what is the value of $a + b$?

A) 2
B) 7
C) 9
D) 11

24.

Ryan and Allison build a ramp to help their elderly cat, Simms, walk up to their bed. They need the ramp to make a 35° angle with their bedroom floor. How long must the ramp be to reach the top of their bed that is exactly three feet off the ground?

A) $\frac{\sin 35^\circ}{3}$
B) $\frac{\sin 55^\circ}{3}$
C) $\frac{3}{\sin 55^\circ}$
D) $\frac{3}{\sin 35^\circ}$

25.

If $x + y = 0$, which of the following must be equivalent to $x - y$?

A) $-2y$
B) $\frac{x}{y}$
C) x
D) x^2

26.

The number of soil samples, s , that Sonal needs for an experiment must be greater than 6 but less than or equal to 13. Which of the following represents an acceptable number of soil samples for Sonal's experiment?

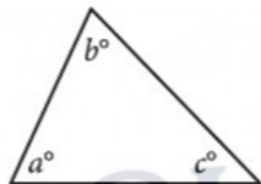
A) $6 < s < 13$
B) $6 \leq s < 13$
C) $6 < s \leq 13$
D) $6 \leq s \leq 13$

27.

If two times a number is equal to that number minus 4, what is the number?

- A) -7
- B) -6
- C) -4
- D) -3

28.



Note: Figure not drawn to scale.

In the triangle above, $a = 34$. What is the value of $b + c$?

29.

The relationship between x and y can be written as $y = mx$, where m is a constant. If $y = 17$ when $x = a$, what is the value of y when $x = 2a$?

30.

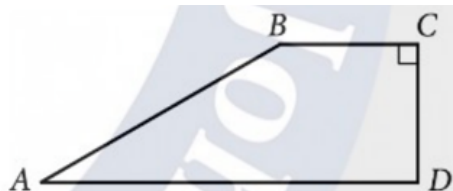
700, 1200, 1600, 2000, x

If the mean of the five numbers above is 1600, what is the value of x ?

31.

Lynne has \$8.00 to spend on apples and oranges. Apples cost \$0.65 each, and oranges cost \$0.75 each. If there is no tax on this purchase and she buys 5 apples, what is the maximum number of whole oranges she can buy?

32.



In quadrilateral $ABCD$ above, $AD \parallel BC$ and

$CD = \frac{1}{2}AB$. What is the measure of angle B ?

- A) 150°
- B) 135°
- C) 120°
- D) 90°

33.

$$H = 1.88L + 32.01$$

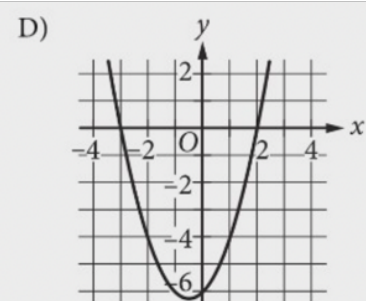
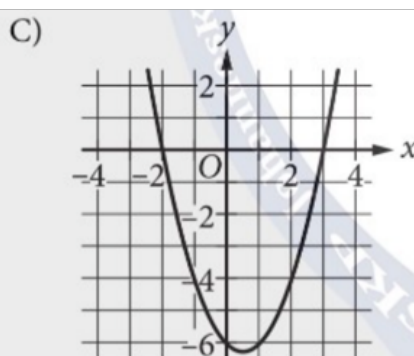
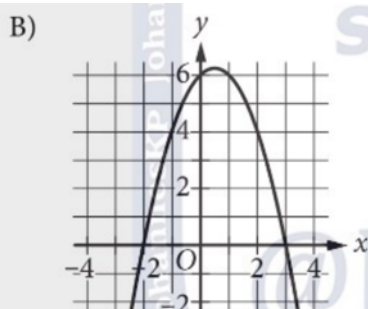
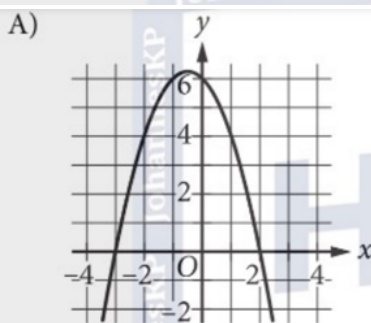
The formula above can be used to approximate the height H , in inches, of an adult male based on the length L , in inches, of his femur. What is the meaning of 1.88 in this context?

- A) The approximate femur length, in inches, for a man with a height of 32.01 inches.
- B) The approximate increase in a man's femur length, in inches, for each increase of 32.01 inches in his height.
- C) The approximate increase in a man's femur length, in inches, for each one-inch increase in his height.
- D) The approximate increase in a man's height, in inches, for each one-inch increase in his femur length.

34.

$$f(x) = (x+3)(x-k)$$

The function f is defined above. If k is a positive integer, which of the following could represent the graph of $y = f(x)$ in the xy -plane?



35.

$$\frac{2}{3}(9x-6)-4=9x-6$$

Based on the equation above, what is the value of $3x-2$?

- A) -4
- B) $-\frac{4}{5}$
- C) $-\frac{2}{3}$
- D) 4

36.

$$2.4x - 1.5y = 0.3$$

$$1.6x + 0.5y = -1.3$$

The system of equations above is graphed in the xy -plane. What is the x -coordinate of the intersection point (x, y) of the system?

- A) -0.5
- B) -0.25
- C) 0.8
- D) 1.75

37.

x	y
a	0
$3a$	$-a$
$5a$	$-2a$

Some values of x and their corresponding values of y are shown in the table above, where a is a constant. If there is a linear relationship between x and y , which of the following equations represents the relationship?

- A) $x + 2y = a$
- B) $x + 2y = 5a$
- C) $2x - y = -5a$
- D) $2x - y = 7a$

38.

The maximum value of a data set consisting of 25 positive integers is 84. A new data set consisting of 26 positive integers is created by including 96 in the original data set. Which of the following measures must be 12 greater for the new data set than for the original data set?

- A) The mean
- B) The median
- C) The range
- D) The standard deviation

39.

$$y = -(x - 3)^2 + a$$

In the equation above, a is a constant. The graph of the equation in the xy -plane is a parabola. Which of the following is true about the parabola?

- A) Its minimum occurs at $(-3, a)$.
- B) Its minimum occurs at $(3, a)$.
- C) Its maximum occurs at $(-3, a)$.
- D) Its maximum occurs at $(3, a)$.

40.

A book was on sale for 40% off its original price. If the sale price of the book was \$18.00, what was the original price of the book? (Assume there is no sales tax.)

- A) \$7.20
- B) \$10.80
- C) \$30.00
- D) \$45.00

41.

$$h(x) = 2^x$$

The function h is defined above. What is $h(5) - h(3)$?

- A) 2
- B) 4
- C) 24
- D) 28

42.

Which of the following is equivalent to $2x(x^2 - 3x)$?

A) $-4x^2$

B) $3x^3 - x^2$

C) $2x^3 - 3x$

D) $2x^3 - 6x^2$

43.

$$f(x) = \frac{x+3}{2}$$

For the function f defined above, what is the value of $f(-1)$?

A) -2

B) -1

C) 1

D) 2

44.

If $20 - x = 15$, what is the value of $3x$?

A) 5

B) 10

C) 15

D) 35